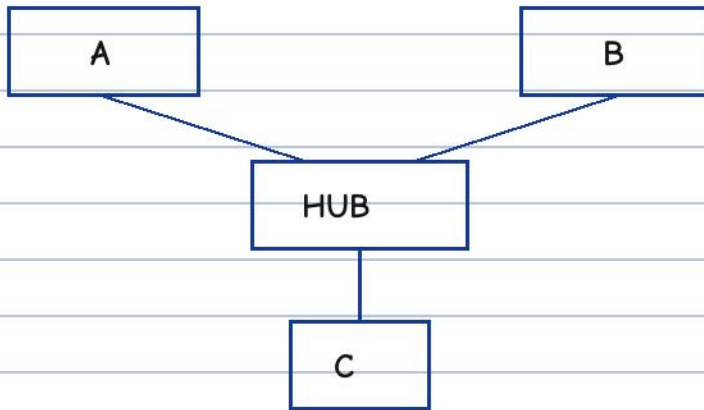


1) Explain behaviour of hub device?

HUB:

- * It is used to connect the multiple network device.
- * It is a broadcast device.

Let us see in step-by-step:



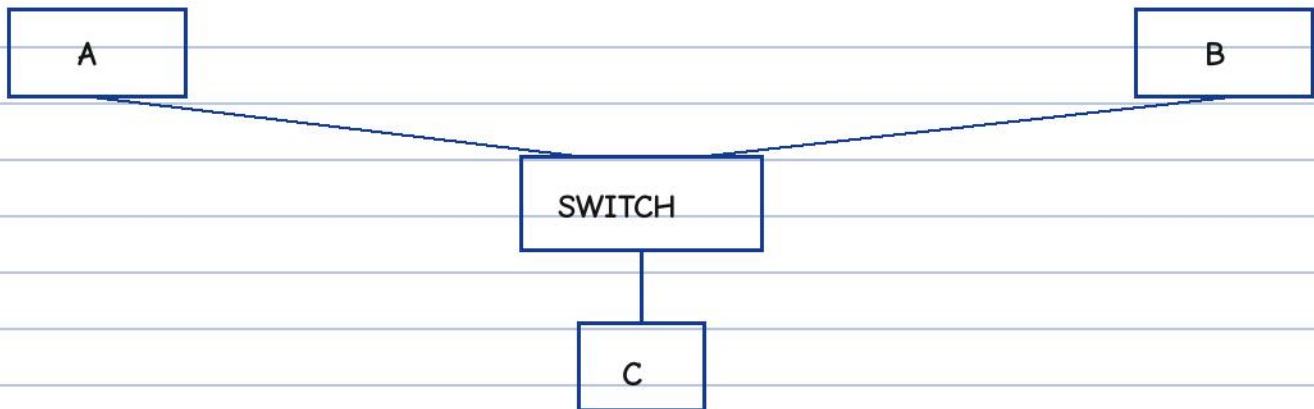
- * We have three PC's, name as A, B and C and port no is A=1, B=2, and C=3.
- * First of all, the data is send to PC A to PC B.

- * Send the data to hub, once the hub is received the data.
- * The data is can be broadcast to all PC.
- * Send for PC B and PC C.
- * PC B receives the data and PC C receive and automatically drop the data.
- * Main think, once the data is shared to the hub, it can be broadcast and collision.
- * Because, of data loss and heavy traffic.
- * **It is called as half duplex.**
- * The hub behaviour is once you shared the data from one PC to another PC, using the hub, the data can be broadcasted.

2) Explain behaviour of switch device?

SWITCH:

- * It is use to connect multiple network device.
- * It contains MAC table.
- * It will act as broadcast + unicast.



- * Shared the data from one PC to another PC.
- * We have three PC = A, B, C.

- * First, you have to assign a IP address to A and B PC.
- * A = 50.0.0.10 and B = 50.0.0.20
- * We have to send the echo data from PC B with the help of switch.
- * The switch behaves, once the data is received and note the port number and MAC address on the MAC table is known as address learning.
- * If first time, you will send the data and stores on the MAC table and broadcast to all PC.
- * Then return from PC B to PC A, it will comes in switch, it will noted the PC B port number and MAC address and check it whether the destination is on the MAC table and directly send to the destination PC.

destination PC.

- * This will called as broadcast + unicast.

3) Explain behaviour of router device?

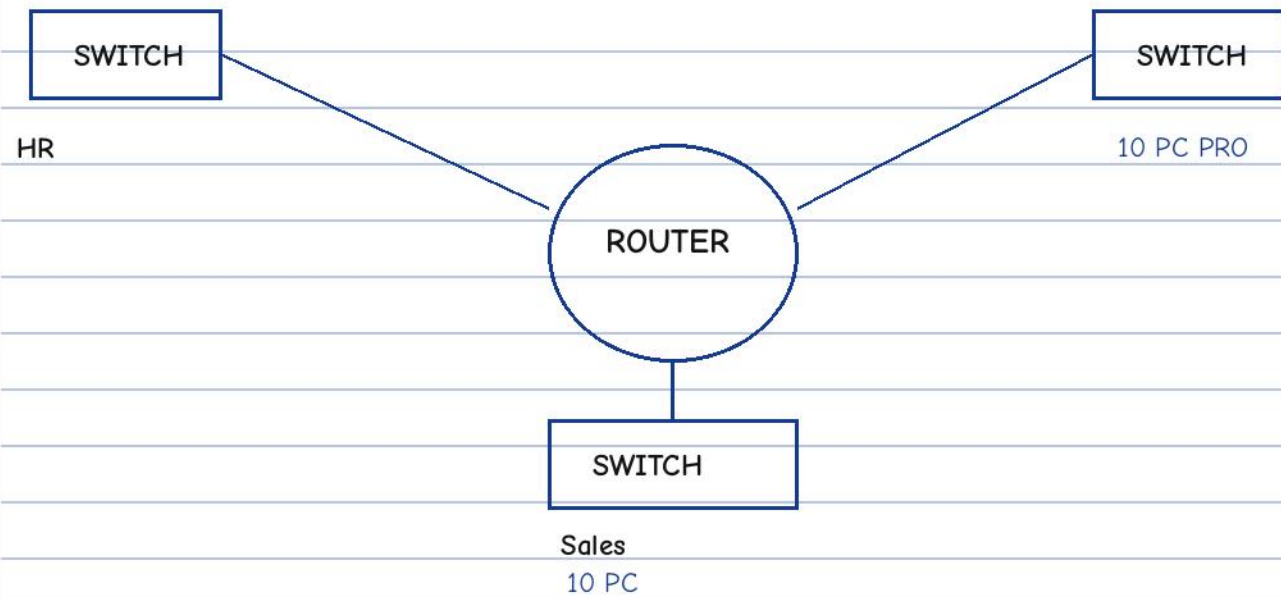
Router:

- * It is use to breaks the broadcast domain.
- * It is a unicast device.
- * It contains routing table.
- * Routing table contains list of rule.

Let's see the below diagram to understand:

210.76.216.11

138.172.18.72



- * We have three department as HR, production, sales.
- * Each department PC have connected by the switch and all the switch can be connected by the router device.
- * The router device performing the data sharing by the way of unicast.

- * Each department have a IP Address and each department have to assign a IP for the router.
- * The router device notes the information of network ID, prefix and interface name in the routing table.
- * Router $\times$ IP $\times$ 3 information.
- * The department to department can be share the data using the router device by the way of unicast.
- * This is called Router device.